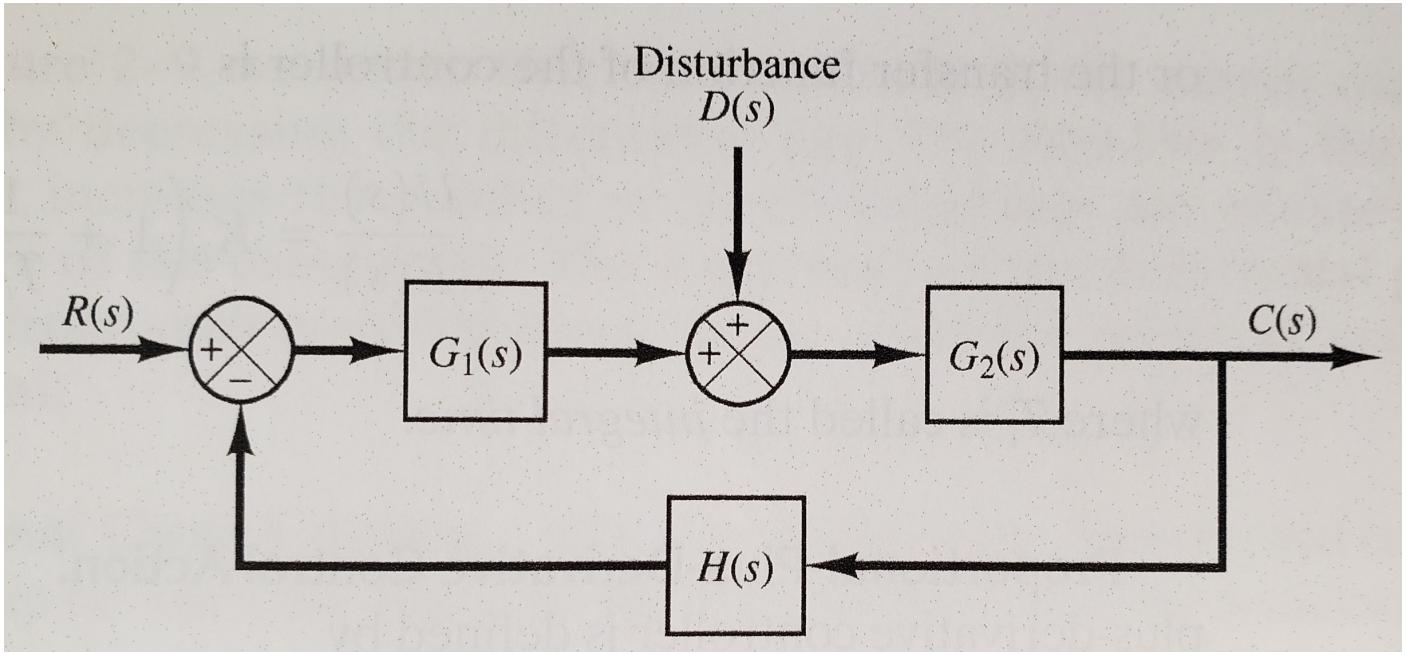
Mechatronics Design Laboratory I

ENME 561/661

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Department of Mechanical and Manufacturing Engineering

2025/09/25

What is the output given input R, disturbance D?

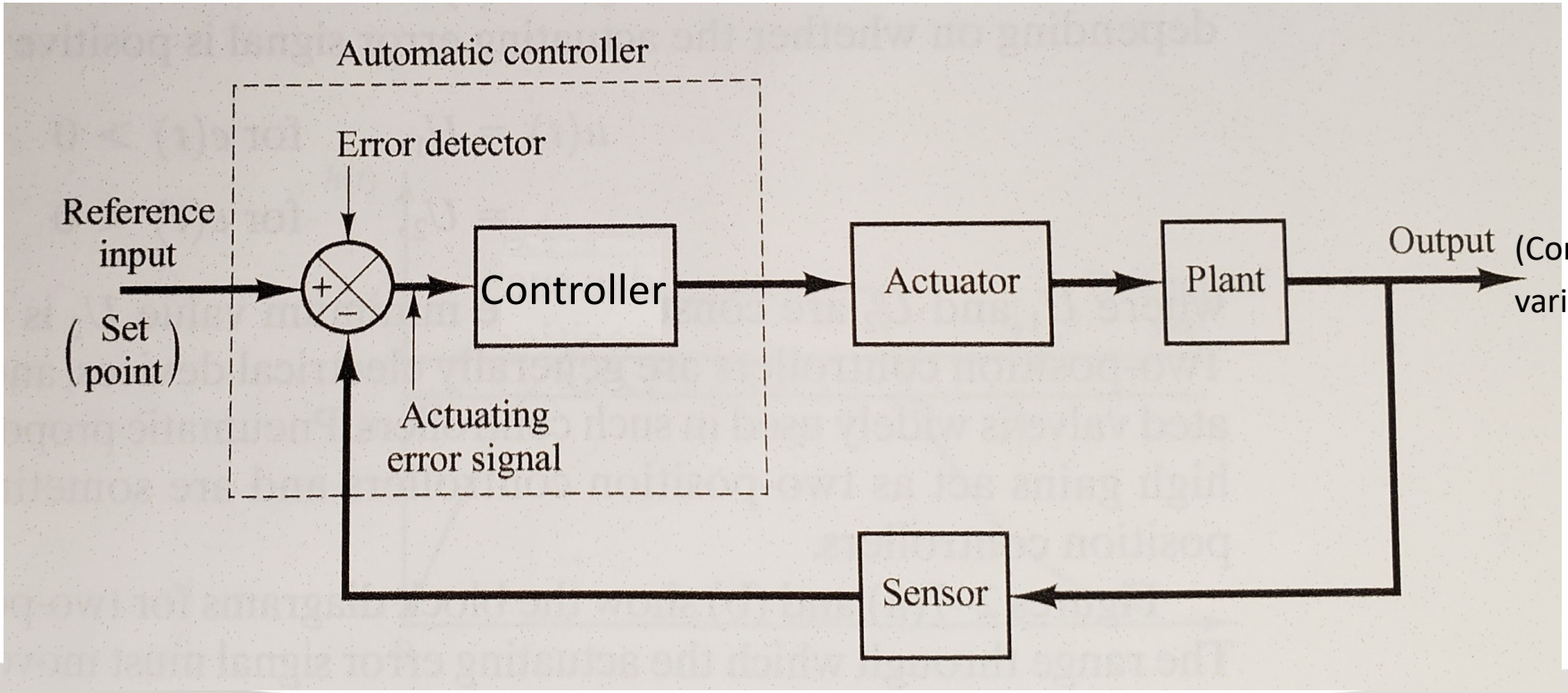


- How to suppress disturbance?

Block diagram of a general controlled system

Goal: Maintain the output at the desired setpoint or follow a reference input.

How: The control output adjusts the action based on the error between the desired setpoint (or reference) and the actual output.



Examples of Industrial Controllers

- On-off (two-position) control action

$$u(t) = U_1, \quad e(t) > 0$$

$$u(t) = U_2, \quad e(t) \leq 0$$

TF=U_1, U_2: two fixed levels

- Proportional control action

$$u(t) = K_p\, e(t)$$

$$TF = U(s) / E(s) = K_p$$

Examples of Industrial Controllers

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- Integral control action
  - $du/dt = K_i\, e(t)$, or $u(t) = K_i \int_0^t e(\tau)\, d\tau$
  - $TF = U(s) / E(s) = K_i / s$
- Proportional derivative control actions (PD)
  - $u(t) = K_p\, e(t) + K_p\, T_d\, de(t)/dt$
  - $TF = U(s) / E(s) = K_p\, (1 + T_d\, s)$
- Proportional Integral Derivative control actions (PID)
  - $u(t) = K_p\, e(t) + (K_p/T_i) \int_0^t e(\tau)\, d\tau + K_p\, T_d\, de(t)/dt$
  - $U(s) / E(s) = K_p\, (1 + 1/(T_i\, s) + T_d\, s)$

More about the Derivate Action

- Derivative kick

- Amplified noise

Arrangement of Actions

- Ideal (Parallel) PID

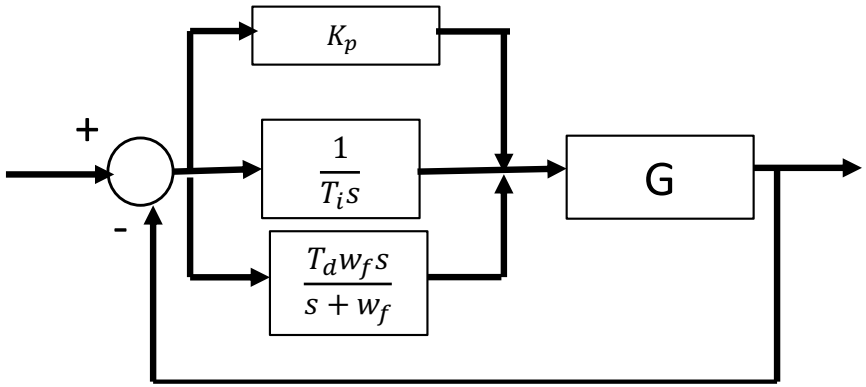
$$C(s) = K_p \left(1 + 1/(T_i\, s) + T_d\, s \right)$$

- Parallel PID (with Filtered D action)

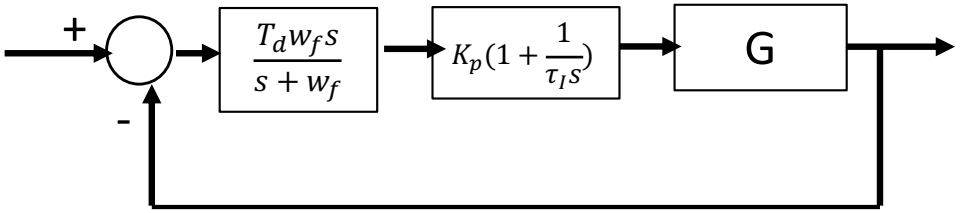
$$C(s) = K_p \left(1 + 1/(T_i\, s) + T_d\, s\, \omega_f / (s + \omega_f) \right)$$

- Series PID (with Filtered D action)

$$C(s) = K_c \left[(T_i\, s + 1) / (T_i\, s) \right] \cdot \left[T_d\, s\, \omega_f / (s + \omega_f) \right]$$

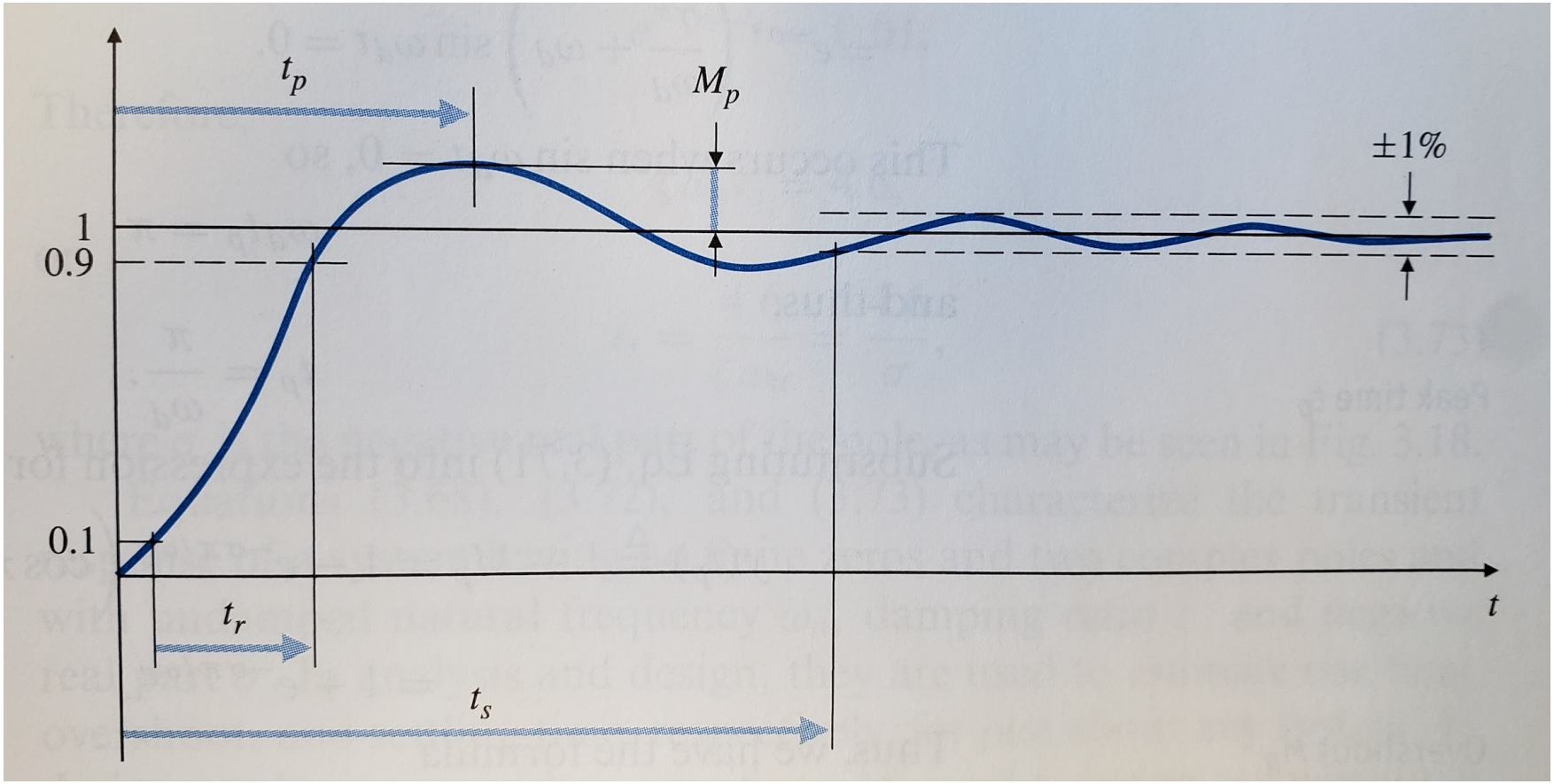


Parallel



Series

Response Characteristics



Reference

- Feedback Control of Dynamic Systems, 8th Edition, Franklin, Powell, Emami-Naeini
- Modern Control Engineering, 5th Edition, Ogata
- Autotuning of PID Controllers: A Relay Feedback Approach, Cheng-Ching Yu
- https://www.mathworks.com